RISE RESEARCH INNOVATION SCHOLARSHIP ENTREPRENEURSHIP

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Subject/Category : Introduction to Databases
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A Data–Driven Food Waste Tracking and Analysis System

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Background, Motivation, and Goals

Food waste is a major global issue that leads to environmental damage and resource waste.
However, there is a lack of ways for individuals to track and understand their own daily food waste behaviors.
This project aims to build a system that can record and analyze the food waste situation at the individual level.

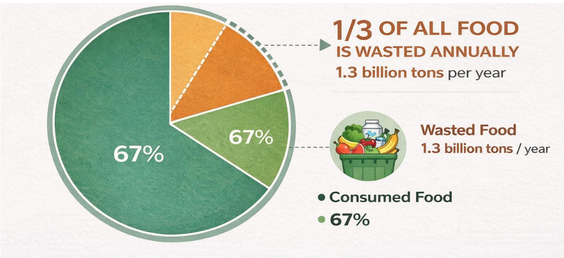


Figure.1 Approximately one-third of global food is wasted annually.

Process, Methods and System

- Built a relational database using MySQL with 5 tables in 3NF
- Implemented foreign key constraints to ensure data integrity
- Developed a web application using Flask
- Designed and executed 17 SQL queries for data analysis
- Enabled full CRUD operations and interactive data visualization

Figure 2: System Workflow Diagram

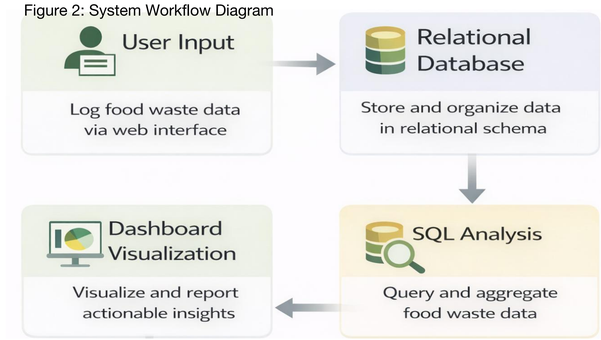
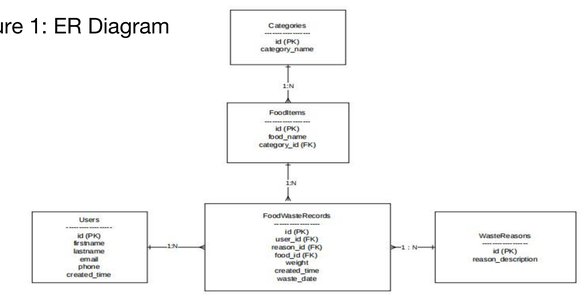


Figure 1: ER Diagram



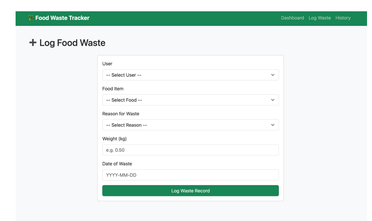


Figure 3: Log Waste Form

Total Waste per User (kg)

User	Total Weight (kg)
Alice Chen	3.50
Carol Nguyen	2.75
Emily Smith	2.75
Bob Martinez	2.65
David Okafor	2.00

Figure 4: User Waste Summary

Category Breakdown

Category	Weight (kg)	% of Total
Prepared Food	3.60	26.4%
Grain	3.50	25.6%
Vegetable	1.55	11.4%
Fruit	1.25	9.2%
Dairy	1.05	7.7%
Meat	1.05	7.7%
Seafood	0.85	6.2%
Beverage	0.80	5.9%

Figure 5: Waste Distribution by Category

Example Queries

```
-- Total waste per category
SELECT category, SUM(weight)
    AS total_weight
FROM WasteRecords
GROUP BY category
ORDER BY total_weight DESC;
```

```
-- Top waste reasons
SELECT reason, COUNT(*) AS count
    FROM WasteRecords
    GROUP BY reason
ORDER BY count DESC;
```

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-- Example Queries
SELECT category, SUM(weight)
    FROM WasteRecords
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```

```
-- Top waste reasons
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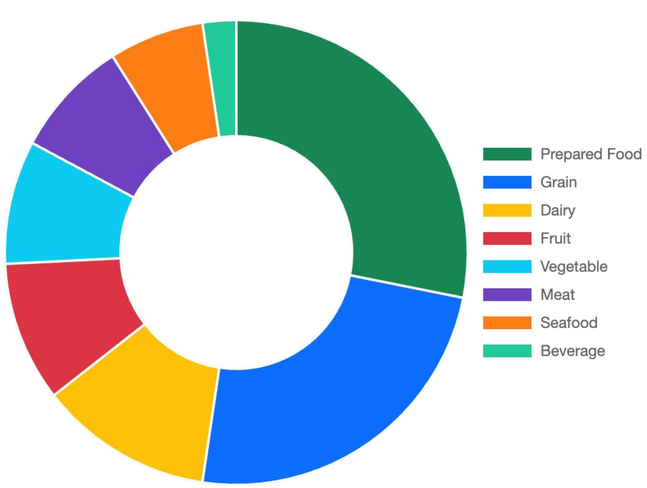
ORDER BY count
ORDER BY count DESC;
```

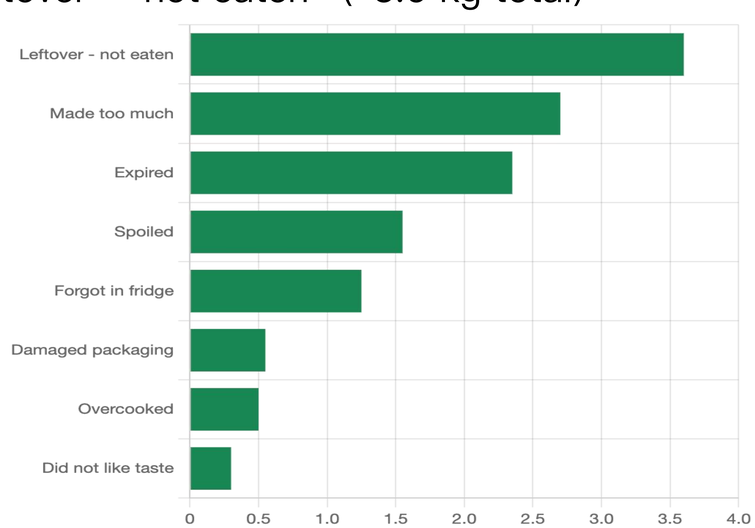
SQL queries are used to aggregate and analyze the data, enabling insights such as total waste per category, most common waste reasons, and temporal trends over time.

The processed data is then visualized through an analytics dashboard, providing users with clear and actionable insights into their consumption behavior.

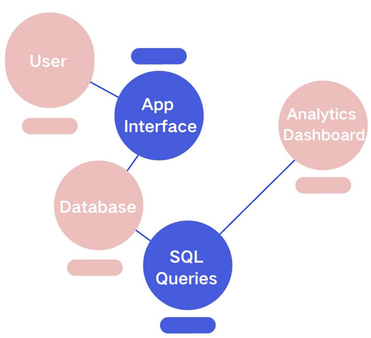
Findings and Products

- Prepared food accounts for 28.1% of total waste
- Grain accounts for 24.2% of total waste
- The most common reason is “Leftover — not eaten” (~3.6 kg total)





Data in this project is user-generated and stored in a relational database system.



Key Insights
- Waste is concentrated in a few key categories
- Behavioral factors dominate waste causes
- Significant variation exists across users
- Consumption patterns influence waste levels

Conclusion and Next Steps

This system demonstrates how database-based applications can provide meaningful insights into real-world behaviors.
By tracking food waste, users can better understand their eating habits and make choices that are more conducive to sustainability.

Future work:
- Expand analytics queries for deeper behavioral insights
- Improve user interface and usability of the web application
- Optimize database performance and scalability